Carbon Permanence Monitoring — End-to-End Checklist

A working to-do list for the four-month build. Tick things off in order; the sequence matters more than the speed.

Week 0 — Before anything else

Do all of this in the first few days. Every item here is a thing that silently blocks you later if skipped.

Accounts (one afternoon)

- [ ] Google Earth Engine — register, create a Cloud project
- [ ] Complete Earth Engine **non-commercial verification** (not instant, start now)
- [ ] NASA Earthdata Login — and authorize applications in profile settings
- [ ] NASA FIRMS — request a free `MAP_KEY`
- [ ] Copernicus Data Space Ecosystem account
- [ ] Copernicus Climate Data Store account — **accept the ERA5 licence** (downloads fail silently otherwise)

Tools

- [ ] Python environment: `earthengine-api`, `geemap`, `pandas`, `scikit-learn`, `shap`, `geopandas`
- [ ] Git repo created, first commit
- [ ] `ee.Authenticate()` runs successfully

Decide and write down

- [ ] Two project types: **forest restoration** (first) and **mangrove** (second)
- [ ] Study sites — see below
- [ ] Study period: suggest 2019–2025 (Sentinel-2 + Dynamic World both available)

Site selection — do not skip this

This is the step that kills projects in Month 3 if rushed.

- [ ] Pick 1 forest site and 1 mangrove site
- [ ] **Confirm each has at least one real disturbance event in the record** (check Hansen `lossyear`, or FIRMS fire history)
- [ ] Check Sentinel-2 coverage isn't destroyed by cloud (count usable scenes per year)
- [ ] Check the mangrove site overlaps Global Mangrove Watch extent
- [ ] Save both boundaries as GeoJSON in the repo

> A perfectly healthy site gives you no positive labels and no case study. You need something to have gone wrong.

Month 1 — Foundation

Week 1 — First real data

- [ ] Write an Earth Engine script: one polygon → monthly median NDVI, 2019–2025
- [ ] Export to CSV, plot the curve
- [ ] **Find the dip that matches your known disturbance**

Stop here until that curve exists. Everything else is theory until it does.

Week 2 — Full feature pipeline for one site

- [ ] Add NDWI (needed for mangrove)
- [ ] Add ERA5-Land: rainfall, temperature, soil moisture
- [ ] Add Dynamic World tree probability
- [ ] Add FIRMS fire counts within 5 km (separate API, not Earth Engine)
- [ ] Add Hansen loss hectares per year
- [ ] Output: one CSV, one row per month, all columns

Week 3 — Baselines and normalization

- [ ] Compute per-site, per-calendar-month baselines (5-year median)
- [ ] Convert every raw value to an anomaly
- [ ] Normalize all anomalies to 0–1
- [ ] Handle the restoration-trend problem (sites are *supposed* to improve)

Week 4 — Config architecture + literature

- [ ] Create `config/forest.yaml` and `config/mangrove.yaml`
- [ ] **Move all indicator lists, weights and thresholds into config — no hardcoding**
- [ ] Read the Verra AFOLU Non-Permanence Risk Tool (v4.2)
- [ ] Read Hansen et al. 2013
- [ ] Write the research-gap section, naming CarbonTrust and Regen Atlas explicitly

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Month 2 — Data at scale + models

Week 5 — Global patch sample

- [ ] Define eligible area (Dynamic World tree cover + Global Mangrove Watch)
- [ ] Generate 300–600 patches of 50–200 ha across multiple countries
- [ ] **Stratified sample** — deliberately oversample patches with Hansen loss

- [ ] Match negatives geographically to positives (same countries/biomes)
- [ ] **Split regions into train / validate / test now. Seal the test set.**

Week 6 — Run the pipeline over everything

- [ ] Batch the feature extraction across all patches
- [ ] Export as tables, not images (watch your EECU quota)
- [ ] Clip to polygons; never `.getInfo()` inside a loop
- [ ] Result: 25k–50k rows of monthly features

Week 7 — Models

- [ ] Define the label: disturbance at t+1 from Hansen `lossyear` / FIRMS / NDVI drop
- [ ] Train Random Forest with **project type as a one-hot feature**
- [ ] Train Isolation Forest **per site**, on that site's own history
- [ ] Spatial cross-validation — train and test on *different regions*

Week 8 — Explainability + experiments

- [ ] SHAP on the Random Forest
- [ ] **Leave-one-type-out experiment**: train forest only, test mangrove
- [ ] Sensitivity analysis: shift weights ±20%, count band changes
- [ ] Freeze models, save to disk

Month 3 — Application

Week 9 — Backend skeleton

- [ ] PostgreSQL + PostGIS running
- [ ] Tables: `projects`, `monthly_features`, `risk_scores`, `alerts`
- [ ] FastAPI: list projects, get one project, get alerts
- [ ] Load your data in

Week 10 — Risk engine

- [ ] Weighted combination driven by the type config
- [ ] Per-project calibration of thresholds (from score history)
- [ ] Escalation override: confirmed fire or large loss → HIGH
- [ ] Hold-back override: anomaly score ordinary → cap at MEDIUM
- [ ] Generate the explanation sentence from SHAP values

Week 11 — Frontend

- [ ] Leaflet map, projects coloured by risk band

- [ ] Project page: level, score, reasons, trend chart, indicator breakdown
- [ ] Alerts list, newest first

Week 12 — Add the second ecosystem

- [ ] Register the mangrove site
- [ ] Run the full chain on it using `mangrove.yaml`
- [ ] Implement auto-detection of project type (Global Mangrove Watch + coastline distance)
- [ ] **Verify nothing needed a code change — only config**

Month 4 — Validation and finalisation

Week 13 — The honest test

- [ ] Open the sealed test regions. Run once.
- [ ] Record precision and recall. **Do not tune afterwards.**
- [ ] Report per-region and per-type breakdowns

Week 14 — Case studies

- [ ] Replay the months before each known disturbance
- [ ] Show risk climbing Low → Medium → event
- [ ] Screenshot the alert naming the right causes

Week 15 — Writing

- [ ] Methods, results, limitations
- [ ] Data sources + licences section (attribution is a condition, not a courtesy)
- [ ] Limitations stated plainly: cloud gaps, coarse climate, Hansen label meaning, no field verification

Week 16 — Delivery

- [ ] Demo script rehearsed — one story, not a feature tour
- [ ] Slides, report, repo cleaned
- [ ] README so someone else can reproduce it

Rules to keep

1. **Config over code.** New ecosystem = new YAML file, never a code change.
2. **Seal the test set.** Open it once, in Month 4.
3. **Never a score without a reason.** Every risk level carries its explanation.
4. **Never claim carbon was lost.** Always "potential risk, field verification recommended."

5. **More training variety, not more tuning.** That's how generalization improves.

Where this usually goes wrong

Risk	Early warning sign	Fix
No disturbance at your sites	Can't find a Hansen loss year	Re-pick sites in Week 0
EECU quota burned	Jobs slow to a crawl	Clip to polygons, export tables
Cloud gaps	Months missing from the NDVI curve	Add Sentinel-1 radar, or pick a drier site
Hardcoded logic	`if type == "mangrove"` in your code	Refactor to config immediately
Month 3 overrun	Backend unfinished at Week 11	Cut dashboard features, never the second ecosystem

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The one thing to do today

Open Earth Engine. Pull a monthly NDVI series for one real polygon. Find the dip.